Project Plan
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1. What scientific questions are you trying to answer with your project?

Overarching question: *Which PBL parameterization schemes best reproduce the nocturnal stable boundary layer over the South Plains region?*

The stable boundary layer is notoriously difficult to model, and the South Plains region has unique features such as a nocturnal low-level jet (LLJ). Further, I'm familiar with several sources of data (200 meter tower at the RTC, and West Texas Mesonet data) that can be used for comparison, making the quantitative analysis approachable.

I'll specifically focus on the accuracy of forecasted near-surface temperature, surface temperature inversion strength, boundary layer wind speed profile, and (if possible) LLJ characteristics.

I'd like to do this in order to learn more about the kinds of PBL closure schemes used in NWP, as I am very interested in boundary layer turbulence. So, in addition to applying the schemes, I'd like to do a deep dive into how each one works, and if possible will spend a couple of minutes of my presentation briefly describing each and how they differ.

2. What simulations are you going to perform?

The ROI is the South Plains region. All simulations will use a nested grid model, with a coarse outer grid covering a significant portion of the continental US and down into Mexico, and a high-resolution (1 or 2 km) inner grid a few times larger than the ROI, centered on Lubbock. The lowest vertical level will be as close to 10 m as possible, and several levels will be included at/below 200 m to allow for a detailed PBL profile for comparison to 200 m tower data.

I'll test at least 3 different PBL schemes. It looks like WRF has over a dozen built-in schemes to choose from, of several different types. So, I'll make sure to choose at least one local TKE scheme and one nonlocal diagnostic scheme, to see how the types compare in usage, function, and accuracy. Other model parameters (region/time period, grid size, vertical levels, other physics parameterizations, etc.) will be held constant across simulations for a fair comparison.

To avoid the results being too case-specific, I'll choose two different periods to run the comparison. The periods will be chosen some time between Nov 2013–Oct 2014, on dates for which I can find both complete GFS data and clean nighttime data from the 200 m tower for comparison (see below). Ideally, I'll find one weakly-stable night with a pronounced LLJ, and one calm strongly-stable night, to see if there are differences in results between the different regimes.

Each simulation will be run over a 20 hour forecast period: 18 UTC (noon CST) to 14 UTC (8 am CST) the next day. The time period of interest is the nighttime, when the SBL is well-developed, so starting at noon should give time for model spinup, and running throughout the entire night will give opportunity to capture the developed SBL, potentially including the nocturnal LLJ (which, when present, typically peaks between midnight and sunrise).

With 3+ schemes run for 2 time periods, this amounts to at least 6 simulations. If time permits, I'll try to test at least one additional scheme.

3. What quantitative analysis are you going to conduct?

I will conduct forecast verification using observations from the following data sources:

- Data from TTU's 200-meter met tower at the RTC
    - For my research, I have spent some time working with data from the tower sourced Nov 2013–Oct 2014.
    - I will compare:
        - Surface heat and momentum fluxes (model outputs vs lowest tower level)
        - TKE vertical profiles (for TKE-based schemes)
        - Wind speed vertical profiles
        - Surface inversion strength: potential temperature difference between lowest tower level/model T2 value and 200 meters
        - Height, strength, and timing of the LLJ (object-oriented approach)
- West Texas Mesonet data from stations in the ROI
    - There are approximately 50 mesonet stations across the South Plains region alone. I will request data from them over the time periods of interest. I have worked with some mesonet data before, so I have some baseline code I can adapt for the analysis.
    - I will compare:
        - Near-surface temperature (mesonet 9 m vs lowest model level)
        - 10 m wind speed

To simplify processing the WRF output data, I'll try and take advantage of the tslist output, specifying the locations of the tower and mesonet stations to have interpolated outputs directly available for analysis. Observation data will be averaged to the same window as model output for direct comparison.

I will compare several metrics of error (RMSE, MAE, ME) for each variable, computed over the nighttime period. For mesonet results, I can visualize the errors over the horizontal region spanned by the stations to see if there is much spatial variation of error. For each case, the results will be compared between the different schemes.

My questions & Dr. Ancell's answers:

- It looks like GFS data might only be available starting in 2015. Is this correct? If so, are there alternative data sources which would be acceptable to use for LBCs?
    - *I'm not sure I know of any sources of GFS data prior to 2015. But, you could use Reanalysis data (either NARR or ERA5, you can search these up to see more about them). Instead of forecasts, these are simply analyses that would force a WRF simulation, which actually makes your experiment cleaner in comparing PBL schemes as it removes some external model forecast error. Both NARR and ERA5 exist over your proposed project dates.*
- Is nesting down to 1 km reasonable? If so, is the 2-grid nesting I described sufficient or should I use a third, intermediate nesting layer to smooth the transition from the coarse initial and boundary data down to the high resolution innermost grid?
    - *Nesting down to 1km is totally fine, it's done all the time with WRF and is appropriate for your setup.*
- From my initial look at the description of WRF's PBL schemes, it seems there might be differences in the kinds of vertical levels they require. Do you have any suggestions on how to handle this while maintaining a fair comparison between the schemes, and making sure it's possible to compare model vs observation values near the surface?
    - *I believe the PBL schemes you will want to use can all use the same vertical level setup. You'll want to design those to go down below 10m I would think, then you can interpolate to the observation locations in the vertical.*

Plan follow-ups and revisions:

- Use ERA5 data instead of GFS for initial & boundary conditions
- Identified events (* key events):
    - 10 Dec 2013 morning *
    - 16 Dec 2013 morning
    - 22 Jan 2014 morning
    - 25 Jan 2014 morning *
    - 14 Feb 2014 morning
- For each identified event, obtain ERA5 and Mesonet data from 18 UTC on the day prior – 18 UTC the day of